

The Literal Constant-Phase Sector at the Strong No-Wrap Prime: Orbit Sparsity and an Endpoint-Scale Literal Bound

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Abstract

TPC-93 exports the complete low-frequency reconstruction window of the matched fixed-shift shell to primitive affine Möbius fibers. TPC-94 proves that every surviving literal phase has conductor exactly 1 or the strong no-wrap prime q_X . The conductor-one branch was left as a potentially full-mass obstruction. We prove that the entire conductor-one sector is already affordable in the literal packet.

The harmless auxiliary prime can be chosen, still with $q_X \asymp Q_X$, larger than every physical row slope, every row difference, and the orbit diameter. A resolved key has phase slope

$$\Omega_\xi = \ell_\xi v_\xi \sigma_\xi B_\xi.$$

Exact common-target provenance gives

$$b_\xi \mid M_\xi - n_\xi, \quad B_\xi \mid b_\xi.$$

The rows are distinct and their difference has modulus below q_X , so $B_\xi < q_X$. The other native factors ℓ_ξ, v_ξ are smaller than q_X , hence on every active literal key

$$q_X \mid \Omega_\xi \iff q_X \mid \sigma_\xi.$$

The opened-divisor relation

$$m_\alpha j + h_0 = u \sigma_\xi$$

then forces $q_X \mid m_\alpha j + h_0$. Since $q_X \nmid m_\alpha$ and the actual orbit interval has diameter less than q_X , each moving row admits at most one such orbit point.

Using the literal row ℓ^1 bounds, the subpower divisor envelopes of both matched prefixes, the exact-content expansion mass, the row-gcd projector mass, and the bounded ℓ^1 mass of the low-window reconstruction multiplier, we obtain

$$|\mathcal{L}_{K,R,X}^{\text{const}}| \ll_{K,h_0,\mathcal{D}} X^{o(1)} Q_X^2.$$

At the critical scales this is $X^{o(1)} N_{0,X}/J_X$, exactly the required amplitude scale. The result removes the complete conductor-one sector from the list of possible polynomial obstructions. It does not bound the remaining short nonconstant, resonant, or high-conductor generic sectors and proves no new signed Möbius cancellation theorem or prime-pair result.

Keywords: fixed shift; affine Möbius fiber; constant additive character; no-wrap prime; orbit sparsity; exceptional sector; endpoint ledger.

Literal-frame convention. Every physical statement retains one prescribed nonzero integer h_0 , both matched polarizations, the actual row coefficients and mask, all content and projector children, every native key, the actual orbit support, and the original global normalization. No collapsed determinant key or rescaled surrogate is substituted for the literal packet.

1 Introduction

On the high- β endpoint row, the physical scales are

$$Q_X = X^{267/400+o(1)}, \quad J_X = X^{133/400+o(1)}, \quad N_{0,X} = J_X Q_X^2.$$

The zero-mode route isolated in TPC-85–TPC-88 asks for the amplitude scale

$$X^{o(1)} Q_X^2 = X^{o(1)} \frac{N_{0,X}}{J_X}. \quad (1)$$

TPC-93 gives an exact low-frequency export of the literal matched shell [4]. After exact-content resolution, one surviving affine key ξ has two primitive forms of determinant h_0 , a finite interval of affine parameters, and additive phase

$$e_{c_\xi q_X}(-r_\xi \varepsilon_\xi \Omega_\xi z), \quad \Omega_\xi = \ell_\xi v_\xi \sigma_\xi B_\xi.$$

TPC-94 proves that literal common-target provenance gives $c_\xi \mid \Omega_\xi$; since q_X is prime and $0 < |r_\xi| < q_X$, the reduced conductor is exactly 1 or q_X [5]. Conductor 1 is equivalent to

$$q_X \mid \Omega_\xi. \quad (2)$$

The reconstruction filter gives this branch its full low-frequency multiplier mass, so Fourier decay alone supplies no discount.

The present paper uses a different feature: the no-wrap modulus is auxiliary. It may be enlarged by a fixed factor without changing a single physical coefficient. Once it dominates all rows, row differences, and orbit diameters, exact common-target provenance makes the progression step divide a nonzero row difference below q_X . The remaining native factors except the complementary divisor σ_ξ are also below q_X . Consequently (2) forces $q_X \mid \sigma_\xi$, and the original orbit target occupies one residue class modulo q_X . The actual orbit interval is shorter than q_X , hence at most one orbit point survives for each moving row.

1.1 What is new

The main result, [theorem 5.2](#), is a literal growing-packet upper bound, not merely a finite-dimensional model. It uses no cancellation of Möbius signs: orbit sparsity alone replaces the usual $O(J_X)$ orbit count by $O(1)$ on the entire conductor-one branch. The two row ℓ^1 bounds then give $X^{o(1)} Q_X^2$.

This is a one-sector closure. In particular:

- the apparent $q_X \mid B_\xi$ alternative is proved absent on active literal support, rather than discarded;
- no estimate is inferred for a nonconstant phase merely from its conductor q_X ;
- the bound has only soft reserve at (1); and
- the determinant compatibility $\lambda_D \leq 2\eta_Z$ and the strict physical budget $\Lambda_{\text{phys}} < 1/400$ remain separate.

1.2 Organization

[Section 2](#) records the exact imported frame and the strengthened no-wrap choice. [Section 3](#) proves the divisibility dichotomy. [Section 4](#) proves the one-orbit-point theorem. [Section 5](#) restores every literal multiplier and proves the endpoint-scale bound. [Section 6](#) describes the exact finite certificate, and [sections 7](#) and [8](#) state what has and has not advanced.

2 The literal frame and a stronger harmless padding

2.1 Rows, orbit points, and opened divisors

Fix $h_0 \in \mathbb{Z} \setminus \{0\}$. On one physical dyadic packet a row is

$$\alpha = (\ell_\alpha, d_\alpha), \quad m_\alpha = \ell_\alpha d_\alpha \asymp_{\mathcal{D}} Q_X,$$

and its orbit target is

$$N_\alpha(j) = m_\alpha j + h_0. \quad (3)$$

The joint orbit support lies in a fixed union of intervals contained in an interval $\mathcal{J}_X$ of diameter $O_{\mathcal{D}}(J_X)$. The two row coefficients satisfy the imported bounds

$$\sum_{\alpha} |\gamma_{\alpha}^{(i)}| \ll_{\mathcal{D}} Q_X, \quad i = 1, 2. \quad (4)$$

These are the literal row coefficients, including prime-source, squarefree-divisor, smooth, and residue factors [2].

Opening the ultra increment on the moving row gives

$$N_\alpha(j) = u\sigma, \quad T_X < u \leq U_{0,X}, \quad (5)$$

where u carries the Möbius–logarithmic ultra coefficient and σ is the complementary divisor. The actual-mask disintegration further introduces a row-gcd divisor $v \mid d_\alpha, d_\gamma$, at most two row-coordinate intervals, and projector children whose total absolute coefficient mass, for each fixed opposite row (one physical column), is

$$X^{o(1)}. \quad (6)$$

The same columnwise statement holds after exchanging the two polarizations. No full-matrix low-rank or projective-mass conclusion is inferred from it.

2.2 Soft divisor envelopes

All effective targets are $X^{O(1)}$. The elementary divisor bound and the explicit matched coefficients therefore give, uniformly on the literal support,

$$\sum_{\substack{u \mid N \\ T_X < u \leq U_{0,X}}} |\mu(u)| \log u \leq X^{o(1)}, \quad (7)$$

$$|H_m(j)| + |A_{m,T_X}(j)| + |A_{m,U_{0,X}}(j)| \leq X^{o(1)}. \quad (8)$$

Every $o(1)$ here is uniform over the fixed dyadic data $\mathcal{D}$. No cancellation is used in (7)–(8).

2.3 The strengthened no-wrap prime

TPC-86 chooses an auxiliary prime $q_X \asymp Q_X$ large enough to embed the determinant support injectively [3]. Since all of the following quantities are $O_{\mathcal{D}}(Q_X)$, Bertrand’s postulate permits the same choice to satisfy

$$q_X > 2 \max_{\alpha} m_{\alpha}, \quad 2 \max_{\alpha, \gamma} |m_{\alpha} - m_{\gamma}|, \quad 2 \max_{\alpha} d_{\alpha}, \quad 2 \operatorname{diam}(\mathcal{J}_X), \quad (9)$$

as well as the original determinant no-wrap inequalities. Enlarging the fixed comparison constant changes neither the physical packet nor any exponent.

Lemma 2.1 (Immediate consequences). *For all sufficiently large X , the choice (9) has*

$$q_X \nmid m_{\alpha}, \quad q_X \nmid \ell_{\alpha}, \quad q_X \nmid v$$

on every active row and row-gcd layer. Moreover any residue class modulo q_X meets $\mathcal{J}_X$ in at most one integer.

Proof. Every displayed positive integer is strictly between 0 and q_X . The final assertion follows because two distinct representatives of one residue class differ by at least q_X , whereas $\text{diam}(\mathcal{J}_X) < q_X$. $\square$

Lemma 2.2 (Literal content steps are below q_X). *Call a child opened-active when its two row targets, primitive masks, and exact-content divisibilities all hold before row-gcd projector reassembly; its parent is allowed to cancel after that reassembly. For every opened-active exact-content child,*

$$1 \leq B_\xi \leq b_\xi < q_X.$$

Proof. Write $b_\xi = c_\xi \kappa_\xi$. This integer divides both physical targets

$$M_\xi j_\xi + h_0, \quad n_\xi j_\xi + h_0.$$

The primitive target provenance imported in TPC-93–TPC-94 gives $(b_\xi, j_\xi) = 1$. Subtracting the targets therefore gives

$$b_\xi \mid (M_\xi - n_\xi)j_\xi \implies b_\xi \mid M_\xi - n_\xi.$$

The physical off-diagonal mask has $M_\xi \neq n_\xi$, while (9) gives $|M_\xi - n_\xi| < q_X$. Hence $b_\xi \leq |M_\xi - n_\xi| < q_X$. Finally $B_\xi = b_\xi / (b_\xi, \sigma_\xi) \leq b_\xi$. $\square$

Remark 2.3. The modulus is still only an auxiliary Fourier-padding device. Lemma 2.1 does not impose a congruence on the original sum and does not select a subsequence of scales.

3 The literal constant-phase equivalence

One soluble exact-content key has

$$b = c\kappa, \quad B = \frac{b}{(b, \sigma)},$$

and phase slope

$$\Omega = \ell v \sigma B. \tag{10}$$

Literal target provenance gives $c \mid \Omega$, and TPC-94 therefore reduces the algebraic phase conductor to 1 or q_X . For every nonzero low frequency, conductor 1 is exactly $q_X \mid \Omega$.

Theorem 3.1 (Literal constant-phase equivalence). *On every opened-active literal key,*

$$q_X \mid \Omega \iff q_X \mid \sigma. \tag{11}$$

Proof. Lemma 2.2 gives $0 < B < q_X$. The prime q_X divides neither ℓ nor v by lemma 2.1. Euclid's lemma applied to $\Omega = \ell v \sigma B$ gives the forward implication in (11); the reverse implication is immediate. $\square$

Corollary 3.2 (No hidden progression-step alternative). *The general algebraic possibility $q_X \mid B$ is empty on opened-active literal support. Thus every opened-active conductor-one key, including a one-point key, retains the target divisibility $q_X \mid \sigma$.*

Remark 3.3 (Algebraic conductor versus interval constancy). On a singleton interval every character is pointwise constant, even when its algebraic conductor is q_X . The sector treated here is the conductor-one sector $q_X \mid \Omega$, not the larger class of all characters whose restriction happens to be constant on one point. Those nonconstant-conductor singletons remain in the short sector.

Remark 3.4 (Why primality matters). For a composite padding modulus, its prime factors could split among ℓ, v, σ, B , and (11) would not follow. The prime no-wrap choice is doing genuine arithmetic work here, although it remains an auxiliary embedding.

4 One orbit point per moving row

The complement σ in (5) is not a free affine slope. It remembers the physical target:

$$m_\alpha j + h_0 = u\sigma.$$

This provenance converts conductor degeneracy into orbit sparsity.

Theorem 4.1 (Orbit sparsity on the constant branch). *Fix a moving physical row α . Among all orbit points $j \in \mathcal{J}_X$, at most one can support a conductor-one key obtained by opening the ultra divisor on that row.*

Proof. By theorem 3.1, a conductor-one key satisfies $q_X \mid \sigma$. The factorization (5) then gives

$$m_\alpha j + h_0 \equiv 0 \pmod{q_X}. \quad (12)$$

Since $q_X \nmid m_\alpha$, this congruence has exactly one residue class

$$j \equiv -h_0 m_\alpha^{-1} \pmod{q_X}.$$

By lemma 2.1, $\mathcal{J}_X$ contains at most one integer in that class. □

Corollary 4.2 (Both polarizations). *For the left-ultra polarization, each left moving row admits at most one constant-sector orbit point while the right row remains arbitrary. For the right-ultra polarization, the symmetric statement holds with the row labels exchanged.*

Remark 4.3 (Divisor choices do not restore the J_X factor). At the unique possible orbit point, several divisors u can make $\sigma = (m_\alpha j + h_0)/u$ divisible by q_X . Their total absolute coefficient is $X^{o(1)}$ by (7). Thus divisor multiplicity is a soft factor; it cannot restore J_X distinct orbit points.

Proposition 4.4 (Sharpness of the one-point statement). *The conclusion of theorem 4.1 cannot be improved to zero using only interval length and primitivity. For any prime $q \nmid m$ and any fixed h_0 , an interval containing $-h_0 m^{-1} \pmod{q}$ has one solution of $q \mid mj + h_0$.*

Proof. Choose the canonical representative of the unique residue class and an interval containing it. No Möbius sign or target mask has been used, so nothing in the stated hypotheses forbids that point. □

5 A fully opened literal mass bound

5.1 Why the moving ultra sum must be opened first

On the left-ultra polarization the signed factor

$$C_{m_\alpha}(j) = \sum_{\substack{u \mid N_\alpha(j) \\ T_X < u \leq U_{0,X}}} a_R(u)$$

contains both conductor-one children

$$q_X \mid \sigma = \frac{N_\alpha(j)}{u}$$

and nonconstant children. Cancellation between those two classes can make $C_{m_\alpha}(j)$ smaller than the conductor-one subsum. Therefore $|C_{m_\alpha}(j)|$ is not an upper bound for the absolute constant-sector child mass. The row-gcd projector has the same issue: its children may cancel even when their absolute constant-sector mass is nonzero.

We consequently take total variation only after every relevant child has been opened. This order also includes projector children whose unexpanded parent is killed by the actual mask; such children cancel in the full projector identity but cannot be silently deleted from a sectorwise absolute majorant.

5.2 Columnwise projector variation

Fix an opposite row $\gamma = (\ell_\gamma, e_\gamma)$. Let $\lambda_{G_X}(v)$ be the exact row-gcd projector coefficient. If the local coprimality indicator is also expanded, its coefficient mass is at most $2^{\omega(v)}$. Put

$$\mathfrak{P}_\gamma = \sum_{v|e_\gamma} |\lambda_{G_X}(v)| 2^{\omega(v)}. \quad (13)$$

The literal TPC-33 divisor bounds give, uniformly in the fixed column,

$$\mathfrak{P}_\gamma \leq X^{o(1)}. \quad (14)$$

For example, on squarefree e_γ , the elementary bounds $|\lambda_{G_X}(v)| \leq \tau(v)$ and $\tau(v) = 2^{\omega(v)}$ give $\mathfrak{P}_\gamma \leq 5^{\omega(e_\gamma)} = X^{o(1)}$.

The point is the order of quantifiers: (14) is used inside the sum over γ . It is never promoted to a low-rank decomposition of the full row matrix.

5.3 The fully opened left majorant

Let

$$\mathcal{U}_q(\alpha, j) = \left\{ u : \begin{array}{l} u \mid N_\alpha(j), \quad T_X < u \leq U_{0,X}, \\ q_X \mid N_\alpha(j)/u \end{array} \right\}.$$

After opening the ultra divisor, row-gcd projector, local coprimality children, exact-content children, finite residue labels, and interval components, the total variation of the left conductor-one sector is bounded by

$$\begin{aligned} \mathcal{V}_L &\leq X^{o(1)} \sum_\gamma |\gamma_\gamma^{(2)}| \mathfrak{P}_\gamma \sum_\alpha |\gamma_\alpha^{(1)}| \sum_{\substack{j \in \mathcal{J}_X \\ q_X \mid N_\alpha(j)}} \\ &\quad \times \left(\sum_{u \in \mathcal{U}_q(\alpha, j)} |a_R(u)| \right) |\mathbf{H}_{m_\gamma}(j)|. \end{aligned} \quad (15)$$

Here the leading soft factor includes:

- the exact-content total variation

$$\sum_{\substack{c|G \\ c \leq C_X}} \sum_{\kappa|G/c} |\mu(\kappa)| \leq \tau(G)^2 = X^{o(1)};$$

- at most two interval components and the finite periodic/residue separation; and
- the total-variation norm of the absolutely convergent Mellin/BV representation.

The finite projective mass and the uniform Mellin/BV total-variation control are precisely the separation statements proved in TPC-25 and TPC-33, uniformly in $(\alpha, \gamma, j, u, v, c, \kappa)$ [1, 2, 4]. Tonelli's theorem applies to this nonnegative majorant. Thus continuous separation variables remain inside their total-variation integral and do not become fictitious finite native keys.

Proposition 5.1 (Fully opened polarization bound). *Each exact polarization has conductor-one child variation*

$$\mathcal{V}_L, \mathcal{V}_R \ll_{\mathcal{D}, h_0} X^{o(1)} Q_X^2.$$

Proof. For each moving row α , the j -sum in (15) contains at most one point by theorem 4.1. At that point,

$$\sum_{u \in \mathcal{U}_q(\alpha, j)} |a_R(u)| \leq \sum_{\substack{u|N_\alpha(j) \\ T_X < u \leq U_{0,X}}} |\mu(u)| \log u \leq X^{o(1)}$$

by (7), while $|H_{m_\gamma}(j)| \leq X^{o(1)}$ by (8). Hence

$$\begin{aligned} \mathcal{V}_L &\leq X^{o(1)} \sum_{\gamma} |\gamma_{\gamma}^{(2)}| \mathfrak{P}_{\gamma} \sum_{\alpha} |\gamma_{\alpha}^{(1)}| \\ &\leq X^{o(1)} \left(\sum_{\gamma} |\gamma_{\gamma}^{(2)}| \right) \left(\sum_{\alpha} |\gamma_{\alpha}^{(1)}| \right) \ll X^{o(1)} Q_X^2 \end{aligned}$$

by (14) and (4). The right-ultra term is identical after exchanging the rows. $\square$

5.4 Restoring the reconstruction frequencies

For fixed $K \geq 2$, the translated low-window reconstruction multiplier obeys

$$\sum_{0 < d_{q_X}(r,0) \leq R_X} |\mathfrak{m}_{K,X}(r)| \ll_K 1. \quad (16)$$

The conductor-one predicate is independent of the nonzero low frequency after the literal divisibility $c_{\xi} \mid \Omega_{\xi}$ has been imposed. The orientation and fixed phase have modulus one. Thus restoring all frequencies costs only (16), not their cardinality.

Theorem 5.2 (Endpoint-scale constant-sector bound). *Let $\mathcal{L}_{K,R,X}^{\text{const}}$ be the sum of all fully opened literal TPC-93 low-window children that*

- (i) *occur in either exact polarization, including children of projector parents that cancel only after reassembly;*
- (ii) *retain every native mask, exact-content label, interval component, prefix, and original coefficient; and*
- (iii) *have algebraic conductor 1, equivalently $q_X \mid \Omega_{\xi}$.*

Then

$$|\mathcal{L}_{K,R,X}^{\text{const}}| \ll_{K,h_0,\emptyset} X^{o(1)} Q_X^2. \quad (17)$$

Proof. The absolute value of the sector sum is at most $\mathcal{V}_L + \mathcal{V}_R$ times the frequency multiplier ℓ^1 mass. Apply [proposition 5.1](#) and (16). $\square$

Remark 5.3 (No quotient estimate). The proof works directly with opened physical children, fixed opposite columns, and their native total variation. Equal determinants, affine forms, phase multipliers, or targets are never merged. Neither signed prefix cancellation nor projector cancellation is used as an upper bound for a selected sector.

6 Exact finite regression

The script `experiments/tpc98_certificate.py` checks the finite divisibility mechanism independently of asymptotic notation. It enumerates nonzero signed shifts, primes, primitive rows, orbit intervals shorter than the prime, opened factorizations $mj + h_0 = u\sigma$, content steps B , and native factors ℓ, v . It verifies:

1. every orbit congruence has at most one point in the interval;
2. literal common-target provenance gives $B \leq b \leq |m - n| < q$;
3. on every active literal key, $q \mid \ell v \sigma B$ is equivalent to $q \mid \sigma$;
4. every enumerated constant key comes from the unique orbit congruence $mj + h_0 \equiv 0 \pmod{q}$; and
5. the absolute exact-content multiplicity is bounded by $\tau(G)^2$.

The regression is deliberately finite. It tests exact logic, signs of h_0 , both row orientations, and consequences of the imported primitive-content provenance; it does not test the asymptotic row ℓ^1 estimate, divisor bounds, or any Möbius cancellation theorem.

7 Endpoint and route ledger

At the critical scales,

$$Q_X^2 = \frac{N_{0,X}}{J_X} = X^{-133/400+o(1)} N_{0,X}.$$

Consequently [theorem 5.2](#) places the entire conductor-one sector exactly at the target amplitude scale. It carries no fixed-power loss, but also supplies no additional fixed-power reserve.

Corollary 7.1 (Removal as a polynomial obstruction). *The conductor-one sector cannot be responsible for a failure of every strict amplitude saving below 133/400. Any remaining polynomial obstruction in the TPC-93 low window must lie in the short sector, the nonconstant q_X -conductor resonant sector, the generic arithmetic sector, or their outer reassembly.*

Proof. For every fixed $\varepsilon > 0$,

$$X^{o(1)} Q_X^2 \ll_{\varepsilon} N_{0,X} X^{-133/400+\varepsilon}.$$

Apply [theorem 5.2](#). □

7.1 Independent ledgers

This sector estimate is not entered twice. In particular:

$$\lambda_D \leq 2\eta_Z, \tag{18}$$

$$\Lambda_{\text{phys}} < \frac{1}{400} \tag{19}$$

remain logically separate requirements. The first relates a future determinant-energy lower bound to zero-mode cancellation. The second counts all other literal fixed-power losses. The soft factor in [\(17\)](#) creates no new fixed exponent in either ledger.

TPC-18 full-block reassembly and primitive generic determinant dispersion also remain independent. Eliminating one exceptional low-frequency sector proves neither.

7.2 The next arithmetic target

The literal conductor gap now leaves a cleaner split:

1. *short fibers*: occupancy must be summed across native keys without replacing the actual weights by unit coefficients;
2. *nonconstant resonant fibers*: the phase has conductor q_X but too little archimedean variation on its own interval;
3. *generic fibers*: a weighted two-form Möbius estimate and coherent outer reassembly are still required.

The next paper should attack the resonant incidence geometry before attempting a generic correlation theorem.

8 Claim boundary

Proved here

- The auxiliary no-wrap prime may be enlarged by a fixed factor, still with $q_X \asymp Q_X$, to dominate every relevant row and orbit diameter.
- Exact common-target provenance gives $B_{\xi} \leq b_{\xi} \leq |M_{\xi} - n_{\xi}| < q_X$, so the apparent $q_X \mid B_{\xi}$ alternative is absent.
- Every literal conductor-one key satisfies $q_X \mid \sigma_{\xi}$, hence $q_X \mid m_{\alpha}j + h_0$.

- Each moving row supports at most one constant-sector orbit point.
- After restoring both polarizations, literal row weights, masks, projector children, exact-content children, and low-frequency multiplier mass, the full conductor-one contribution is $O(X^{o(1)}Q_X^2)$.

The finite divisibility statements are L0. Their exact attachment and the growing literal exceptional-sector upper bound are L1 progress. The proof is absolute and uses no new signed-correlation estimate.

Not proved here

- No bound is proved for the complete short sector or the nonconstant resonant and generic sectors.
- No natural post-bin mass or determinant-energy lower bound is proved.
- No fixed-form two-point Möbius theorem, parity breakthrough, Hardy–Littlewood asymptotic, prime-pair lower bound, or twin-prime conclusion is proved or implied.
- Nothing is specialized to $h_0 = 2$. The shift remains one prescribed nonzero integer throughout.

Thus this paper closes one literal exceptional branch but does not claim L2 progress on the full growing fixed-shift coefficient.

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